DAY 1 · THURSDAY 22 OCTOBER

Welcome to ELELEM Chile

From LLMs to Agentic AI — three days to understand, not just do

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Harvard University

THE IDEA

Understand, not just do

- Anyone can wire up an agent in ten lines of code. The hard part is knowing **why it works — and why it breaks**.
- So we go under the hood: how a language model predicts, what retrieval really adds, what turns an LLM into an agent, and how these systems fail.
- Every lecture is followed by a **lab** where you see the same idea in code.
- Ask questions at any time — interrupting is welcome.

You will leave able to

- Explain what an LLM does — and what it cannot do on its own
- Build a RAG pipeline and improve its retrieval
- Build tool-using agents, and know when one agent is enough
- Spot the risks, and design an evaluation for your own system

THREE DAYS

From a language model to agentic systems

Thursday · LLMs → RAG

- Language models: from n-grams to ChatGPT
- ChatGPT lab
- Retrieval-augmented generation
- Industry talk · RAG lab

Friday · Agents

- Capstone brief — form teams
- Agents I: the loop, tools, MCP · lab
- Agents II: context, harnesses, skills · lab
- Industry talk · capstone worked example

Saturday · Landscape

- The AI landscape & multi-agent systems
- Limitations, risks, evaluation
- Capstone reviews and discussion

Capstone: in teams, you design an agentic system — architecture and technical design, no code. Brief on Friday 9:00, designs due Saturday 9:00, reviews Saturday 11:20.

TODAY · THURSDAY 22 OCTOBER

The plan for today

Time	Session
9:00–10:30	Language models I — what a language model is, neural LMs, attention
10:30–10:50	Coffee
10:50–12:30	Language models II — transformers, BERT and GPT, from GPT to ChatGPT
12:30–13:30	Lunch
13:30–15:10	Lab · ChatGPT — from a base model to an assistant
15:10–15:30	Coffee
15:30–17:00	Retrieval-augmented generation
17:00–17:30	Industry talk · Ignacio Becker, CMPC
17:30–18:00	Lab · RAG

THE TEAM

Who is who

Pavlos Protopapas

Instructor

Harvard University

Guillermo Cabrera-Vives

Instructor

Universidad de Concepción

Mabel Vidal

Teaching team

Universidad de Concepción

Matías Olivas

Teaching assistant

Labs

Karla Cerda

Organizer

Executive Director, CDIA

Valentina Riveros

Support team

Logistics

Guest speakers: Ignacio Becker (CMPC) on Thursday · a Lead ML Engineer from CDIA on Friday

PRACTICALITIES

Good to know

Wi-Fi	[network — to fill in] · password [password — to fill in]
Materials	pavlosprotopapas.github.io/elelem-chile-2026 — slides and Colab notebooks, posted before each session
Breaks	Coffee 10:30 and 15:10 · lunch 12:30–13:30
Logistics	Karla Cerda and Valentina Riveros (CDIA) · cdia@inf.udec.cl
Language	Lectures and materials in English — questions in Spanish are fine

Scan for the workshop page

BEFORE THE FIRST LAB (13:30)

Run the readiness check

1

Open the check

Workshop page → Before you arrive → Open in Colab

2

Save a copy

File → Save a copy in Drive

3

Add your key

🔑 Secrets → OPENAI_API_KEY (about USD 10 of credit)

4

Run all cells

The last line prints ✅ ready

No ✅ ready? Bring the output to any of us at the coffee break — before the first lab.